Operating Systems Project Report

CPU Priority Scheduling Simulation

Group 10 — Course Code: 012012500105

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Repository: https://github.com/Hhanguu/PriorityScheduling
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1. Introduction

1.1 Background

CPU scheduling is a fundamental mechanism in operating systems that determines which process gets CPU time and for how long. Priority scheduling assigns a numeric priority to each process; the CPU is always given to the highest-priority ready process (lowest priority number = highest priority).

This project implements and compares two variants of priority scheduling:

- **Non-preemptive Priority Scheduling** — once a process is selected to run, it executes until it finishes, regardless of new arrivals.
- **Preemptive Priority Scheduling (SRPT-style)** — if a newly arrived process has strictly higher priority than the running one, it preempts the CPU immediately.

1.2 Objectives

1. Implement both algorithms correctly and efficiently in C++17.
2. Build a Qt6 desktop GUI that visualizes results as a Gantt chart and comparison table.
3. Read input from CSV; export five result CSV files per run.
4. Measure performance up to 2,000,000 processes and confirm O(n log n) behavior.
5. Cover correctness with an automated test suite (unit + integration + GUI).

1.3 Team

#	Full Name	Student ID	Responsibility
1	Hưng (Leader)	—	Architecture, CMake, integration tests, benchmark, GUI review
2	Kiệt NP	—	NonPreemptiveScheduler, N01–N18 unit tests
3	Vĩ P	—	PreemptiveScheduler, P01–P18 unit tests
4	Minh Hưng GUI	—	MainWindow, GanttChart, ProcessTableModel, GUI tests
5	Lan CSV	—	CsvHandler, C01–C25 unit tests

2. Problem Analysis

2.1 Input and Output Specification

Input: CSV file with header `pid,arrival,burst,priority`

```
pid,arrival,burst,priority
1,0,5,3
2,1,3,2
3,2,8,1
```

Constraints (enforced by `CsvHandler`):

Field	Rule
`pid`	Integer > 0, unique within file
`arrival`	Long long ≥ 0
`burst`	Long long > 0
`priority`	Integer, lower = higher priority

Output: Five CSV files per run (when both algorithms are selected):

File	Content
`np_result.csv`	CT, TAT, WT, RT for each process (non-preemptive)
`np_gantt.csv`	Gantt segments: pid, start, end
`p_result.csv`	CT, TAT, WT, RT (preemptive)
`p_gantt.csv`	Gantt segments (preemptive)
`comparison.csv`	Average WT, TAT, RT side by side

2.2 Computed Metrics

For each process i:

```
CT[i]  = Completion Time
TAT[i] = CT[i] - AT[i]           (Turnaround Time)
WT[i]  = TAT[i] - BT[i]          (Waiting Time)
RT[i]  = first_start[i] - AT[i]  (Response Time)
```

Average values are computed by `Metrics::fillAverages` after scheduling.

2.3 Tie-Breaking Rule

When multiple processes share the same priority: **lower arrival time wins**. When arrival times are equal: **earlier position in the input CSV wins**. This rule applies to both algorithms and is tested explicitly.

2.4 Key Challenges

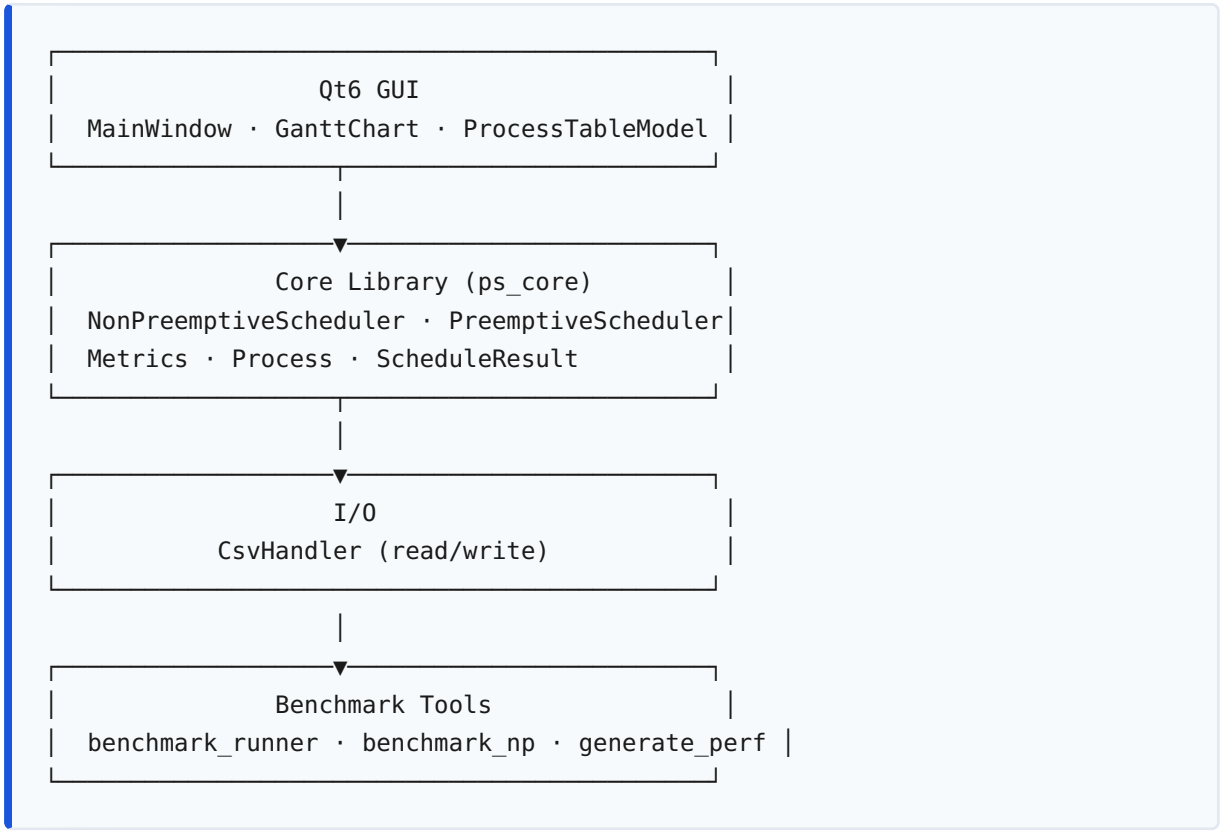
Challenge	Solution
Non-preemptive $O(n^2)$ with naive list scan	Min-heap + FIFO per priority level → O(n log n)
Preemptive tick-by-tick simulation is O(n × T)	Event-based simulation, jump to next event → O(n log n)
GUI table with 1,000,000 rows freezes	Virtual `QTableView` + `ProcessTableModel` (only render visible rows)
Gantt chart unreadable with many segments	`QScrollArea` + minimum 24 px per segment

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3. System Design

3.1 Architecture

The system is divided into four independent layers:



Dependency rule: `core` does not include Qt or I/O. `io` does not include Qt. `ui` may include all. This keeps both schedulers testable without Qt.

3.2 Core Data Structures

```
struct Process {
    int       pid;           // > 0, unique
    long long arrivalTime;   // AT >= 0
    long long burstTime;     // BT > 0
    int       priority;      // lower = higher priority
};

struct GanttSegment {
    int       pid;           // -1 = IDLE
    long long startTime;
    long long endTime;
};

struct ScheduleResult {
    std::vector<ProcessResult> processes;  // same order as input
    std::vector<GanttSegment>  gantt;
    double averageWaitingTime    = 0.0;
    double averageTurnaroundTime = 0.0;
    double averageResponseTime   = 0.0;
};
```

All time fields use `long long` to support arrival times up to 10^{12}.

3.3 Module Responsibilities

Module	Class / Target	Responsibility
`src/core/`	`NonPreemptiveScheduler`	Schedule without preemption
`src/core/`	`PreemptiveScheduler`	Schedule with preemption
`src/core/`	`Metrics`	Compute and fill averages
`src/io/`	`CsvHandler`	Read/validate CSV, write 5 output files
`src/ui/`	`MainWindow`	UI orchestration
`src/ui/`	`GanttChart`	Paint Gantt segments inside `QScrollArea`
`src/ui/`	`ProcessTableModel`	Virtual model for `QTableView`
`src/benchmark/`	`Benchmark`	Shared benchmark utilities (timing, stats, RSS wrapper)
`src/benchmark_main.cpp`	`benchmark_runner`	Measure PreemptiveScheduler
`tools/benchmark_np.cpp`	`benchmark_np`	Measure NonPreemptiveScheduler (5 runs)
`tools/generate_perf.cpp`	`generate_perf`	Generate benchmark CSV

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4. Implementation

4.1 Non-preemptive Priority Scheduler

Algorithm (pseudocode):

```
function schedule(processes):
    sort indices by (AT, inputIndex)
    ready = min-heap ordered by (priority, AT, inputIndex)
    time = 0

    while uncompleted processes remain:
        if ready is empty:
            append IDLE segment [time, next_arrival]
            time = next_arrival

        load all processes where AT <= time into ready
        i = ready.pop()                    // highest priority
        start = time
        time += BT[i]
        record CT, TAT, WT, RT for i
        append Gantt segment [start, time]
```

Key implementation decisions:

- Indices sorted by `(AT, inputIndex)` using `std::sort` with explicit tie-break key — no `stable_sort` needed.
- Ready queue uses `std::map<int, FIFO>` keyed by priority: `map` gives the minimum priority in O(log k), where k is the number of distinct ready priorities (typically small). Each priority level uses an intrusive linked array for O(1) FIFO access.
- This reduces selection cost compared to a global heap over all waiting processes.

Complexity: O(n log n + n log k), where k ≤ distinct priorities.
Memory: O(n).

4.2 Preemptive Priority Scheduler

Algorithm (pseudocode):

```
function schedule(processes):
    remaining[i] = BT[i] for all i
    start[i] = -1 for all i
    sort indices by AT (stable)
    ready = min-heap ordered by (priority, AT, inputIndex)
    time = 0

    while completed < n:
        load all arrivals with AT <= time into ready
        if ready is empty:
            jump to next arrival (IDLE segment)
            continue

        curr = ready.pop()
        eventTime = time + remaining[curr]

        for each upcoming arrival a with AT < eventTime:
            if priority[a] < priority[curr]:
                eventTime = AT[a]        // preempt at exact arrival
                break
            else:
                push a into ready        // higher number = lower priority, no
preempt

        remaining[curr] -= (eventTime - time)
        append/merge Gantt segment [time, eventTime]
        time = eventTime

        if remaining[curr] == 0:
            record CT, TAT, WT, RT
            completed++
        else:
            push curr back into ready   // preempted
```

Key implementation decisions:

- **Event-based simulation** (not tick-by-tick): the scheduler jumps directly to the next significant time point (next arrival or next burst completion). This eliminates the O(T) dependency on total burst time.
- **Arrival at exact eventTime boundary** uses strict `<` (not `≤`) so a process arriving exactly when the current one finishes is loaded in the next iteration — preventing zero-length Gantt segments.
- Adjacent same-PID Gantt segments are merged during emission.

Complexity: O(n log n + g), where g = number of Gantt segments.
Memory: O(n + g).

4.3 CSV Handler

`CsvHandler::readProcesses` validates:

- Header must be exactly `pid,arrival,burst,priority`
- Each field must parse as an integer
- `pid > 0` and unique; `burst > 0`; `arrival >= 0`
- `pid`, `priority` within `int` range; `arrival`, `burst` within `long long` range

Error collection: all invalid lines are collected (up to 100 errors) rather than stopping at the first. The error string is returned to the caller; `MainWindow` truncates to 10 lines before display.

4.4 GUI

MainWindow flow:

```
Open CSV → CsvHandler::readProcesses → display tableInput
Run → NonPreemptiveScheduler and/or PreemptiveScheduler
    → Metrics::fillAverages → update tableResult + averages + GanttChart
Export → CsvHandler::write* → output/
```

Virtual table: `tableInput` and `tableResult` use `QTableView` backed by `ProcessTableModel`, a `QAbstractTableModel` that generates cell strings on demand. It holds a pointer to the existing `std::vector<Process>` / `ScheduleResult` without copying — reset cost is O(1), display cost is proportional to visible rows only.

Gantt chart scrolling: `GanttChart` is wrapped in a `QScrollArea` (horizontal scroll only). The widget's minimum width is `segments × 24 px`, capped at 1,000,000 px (Qt's widget size limit is 16,777,215 px). When this exceeds the frame width, the scroll bar appears. The `paintEvent` only draws segments intersecting `event->rect()`, so scroll performance is independent of total Gantt width.

Run mode: the `comboMode` selector allows running Non-preemptive only, Preemptive only, or both with comparison.

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5. Testing

5.1 Test Framework

All test suites use **doctest** (single-header, `tests/third_party/doctest.h`). Tests are compiled into three separate executables and registered with CTest.

```
cmake -S . -B build
cmake --build build
QT_QPA_PLATFORM=offscreen ctest --test-dir build --output-on-failure
```

5.2 Test Summary

Suite	Executable	Test Cases	Assertions	Scope
Unit	`unit_tests`	79	112,207	Schedulers, CSV, Metrics
Integration	`integration_tests`	10	318	Full pipeline: CSV → schedule → export
GUI	`gui_tests`	10	215	Headless Qt (offscreen)
Total	**ctest 3/3**	**99**	**112,740**	

5.3 Selected Test Cases

Non-preemptive (N01–N18):

ID	Description	Result
N01	Standard 10-process sample	Full Gantt + averages match expected CSV
N02	IDLE at start (first arrival > 0)	Correct IDLE segment
N04	Tie: same priority, same arrival	FIFO by input order
N09	Input with 2,000 processes	Correct output, < 1 s
N13	40 processes, all AT=0 — tie-break by index	Catches pid-based tie-break bug
N14	Process arrives exactly when CPU frees	No zero-length IDLE
N16	Priority INT_MIN and INT_MAX	No overflow in comparison
N17	IDLE segment lasting 10^{12} time units	No `long long` overflow

Preemptive (P01–P18):

ID	Description	Result
P01	Standard sample	Gantt includes preemption segments
P05	Arrival exactly at event boundary	No zero-length Gantt segment
P10	Same priority — no preemption	Running process completes
P15	30,000 processes	Correct output, < 2 s
P18	Stress: 500 processes random	Full Gantt coverage, correct averages

CSV (C01–C25):

ID	Description
C01	Valid 10-process file → 10 processes loaded
C03	Missing header → error message, 0 processes
C15	Multiple errors in one file → all errors reported
C22	500-process file → all loaded correctly

GUI (U01–U16):

ID	Description
U01	Load `gui/sample.csv` → table displays all rows in order
U03	Run NP → Gantt matches `ket_qua/gui/sample_np_gantt.csv`
U09	Export both → 5 files produced
U14	Load 10,005-row file → table shows all rows; Gantt scroll bar appears
U16	Load `sample.csv` (10 segments) → no scroll bar

Integration (I01–I10):

ID	Description
I01	Full pipeline: `sample.csv` → NP → result matches expected CSV
I02	Full pipeline: `sample.csv` → P → result matches expected CSV
I07	2,000-process stress: NP produces 2,001 Gantt segments, correct span

5.4 Correctness Strategy

- Preemptive results verified against an independent $O(n^2)$ oracle on 200 random inputs (fixed seed).
- Gantt continuity invariant (`[0, max(CT)]`, no gaps, no overlaps) checked in every scheduler test.
- Mutation testing on GUI result table: forcing the wrong row index turns 27 assertions red.

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6. Performance Benchmarks

6.1 Environment

Item	Value
Machine	Intel Core i7-1185G7 @ 3.00 GHz (developer's Windows machine)
Verification machine	Intel Core i5-11400F, 15 GiB RAM, Ubuntu 24.04
Compiler	GNU C++ 16.2.0 / GCC 13.3.0
Build type	Release (`-O2 -DNDEBUG`)
Measurement	`std::chrono::steady_clock`, scheduler only (excludes CSV I/O)

6.2 Non-preemptive Scheduler

Measured with `benchmark_np <file> 5` (5 runs, mean and median reported).

Processes	Mean (ms)	Median (ms)	Peak RSS (MiB)
100,000	13.7	13.7	17
1,000,000	242	243	106
2,000,000	577	578	255

Optimization applied: replaced naive list scan ($O(n^2)$) with min-heap + FIFO-per-priority-level ($O(n \log n + n \log k)$). Speedup on 1M processes: ~3.4× (1,180 ms → 353 ms on developer machine; 242 ms on verification machine).

Targets set in `docs/NonPreemptive.md`: **NP 1M < 400 ms, 2M < 1,000 ms** — both met.

6.3 Preemptive Scheduler

Measured with `benchmark_runner <file>` (single run).

Processes	Time (ms)
100,000	27
200,000	64
500,000	242
1,000,000	572

Optimization: replaced tick-by-tick simulation (O(n × T) where T = total burst time, potentially billions) with event-based simulation (O(n log n)). The old implementation took 311 seconds for 100,000 processes; the new one takes 27 ms — a ~11,500× speedup.

6.4 GUI with 1,000,000 Rows

Measured on i5-11400F, offscreen 1180×900, Debug build.

Operation	Time
`loadCsv` (read + parse)	272 ms
`runSchedules` (NP only)	251 ms
`runSchedules` (both NP + P)	835 ms
Window grab (screenshot)	21 ms
Scroll to last row	1 ms
Peak RSS	247 MiB

The virtual table model and event-based Gantt painting make 1,000,000-row display feasible without a worker thread.

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7. Results and Discussion

7.1 Sample Run — 10 Processes

Input (`input/sample.csv`):

PID	AT	BT	Priority
1	0	5	3
2	1	3	2
3	2	8	1
4	3	2	4
5	4	4	5
6	5	1	1
7	6	3	3
8	7	2	2
9	8	5	1
10	9	2	4

Non-preemptive Gantt:

`P1[0–5] → P3[5–13] → P6[13–14] → P2[14–17] → P4[17–19] → P8[19–21] → P10[21–23] → P7[23–26] → P5[26–30] → P9[30–35]`

Metric	NP	P
Avg Waiting Time	8.00	7.80
Avg Turnaround Time	11.50	11.30
Avg Response Time	8.00	4.70

Observation: Preemptive scheduling achieves lower average waiting and response time because high-priority processes that arrive late immediately preempt lower-priority ones. However, it generates more Gantt segments and is slightly slower to compute due to additional heap operations.

7.2 Algorithm Comparison

Property	Non-preemptive	Preemptive
Preemption	None	Immediate on higher-priority arrival
Response time	Higher (must wait for running process)	Lower (gets CPU immediately)
Starvation risk	Yes (low-priority processes may wait indefinitely)	Yes (more severe — can be interrupted repeatedly)
Complexity	O(n log n)	O(n log n)
Gantt segments	At most n + IDLE gaps	Up to $O(n^2)$ in worst case
Implementation	Simpler	Event-based, more complex

7.3 Limitations

- Both algorithms are susceptible to starvation of low-priority processes. No aging mechanism is implemented (out of scope for this project).
- Priority values are plain integers; no support for dynamic priority adjustment.
- Benchmark data is generated with a fixed random seed; real-world workloads may differ.

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8. Conclusion

This project successfully implements and compares Non-preemptive and Preemptive Priority CPU scheduling:

1. **Correctness** is validated by 99 test cases and 112,740 assertions across unit, integration, and GUI test suites. All 3 ctest suites pass on both Debug and Release builds.
2. **Performance** meets the stated targets: NP handles 1,000,000 processes in 242 ms and 2,000,000 in 577 ms; P handles 1,000,000 in 572 ms. Both achieve O(n log n) behavior through min-heap + event-based design.
3. **Usability** is addressed by the Qt6 GUI: virtual table display for unlimited row counts, horizontal Gantt scroll, one-click export of 5 CSV files, and clear error messages for invalid input.

4. **Maintainability** is supported by a frozen API contract (`docs/api.md`), a strict dependency rule (core ← io ← ui), and comprehensive documentation in `docs/` .

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9. Member Contributions

Member	Module	Deliverables
Hưng (Leader)	Architecture, build, integration	`CMakeLists.txt` , `docs/api.md` , `docs/conventions.md` , integration tests (I01–I10), benchmark tools (`benchmark_np` , `generate_perf`), `report/non-preemptive-optimization.md`
Kiệt NP	Non-preemptive scheduler	`src/core/NonPreemptiveScheduler.{h,cpp}` , unit tests N01–N18 (`np_regression_test.cpp` , `np_edge_test.cpp`)
Vĩ P	Preemptive scheduler	`src/core/PreemptiveScheduler.{h,cpp}` , unit tests P01–P18 (`preemptive_test.cpp`)
Minh Hưng GUI	Qt GUI	`src/ui/MainWindow.{h,cpp}` , `src/ui/GanttChart.{h,cpp}` , `src/ui/ProcessTableModel.{h,cpp}` , GUI tests U01–U16 (`gui_test.cpp`)
Lan CSV	CSV I/O	`src/io/CsvHandler.{h,cpp}` , unit tests C01–C25 (`csv_handler_test.cpp`)

Appendix A — Build Instructions

```
# Linux
sudo apt install qt6-base-dev cmake g++ git
git clone https://github.com/Hhanguu/PriorityScheduling.git
cd PriorityScheduling
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build -j4

# Run application
./build/PriorityScheduling

# Run all tests
QT_QPA_PLATFORM=offscreen ctest --test-dir build --output-on-failure

# Generate benchmark data and measure
./build/generate_perf --out input --sizes 100000 1000000
./build/benchmark_runner input/perf_100000.csv
./build/benchmark_np input/perf_1000000.csv 5
```

Appendix B — Repository Structure

```
src/
  core/      Process.h, GanttSegment.h, ScheduleResult.h, Metrics.h
             NonPreemptiveScheduler.{h,cpp}, PreemptiveScheduler.{h,cpp}
  io/        CsvHandler.{h,cpp}
  ui/        MainWindow.{h,cpp}, GanttChart.{h,cpp}, ProcessTableModel.{h,cpp}
  benchmark/ Benchmark.{h,cpp}
  main.cpp
  benchmark_main.cpp
tools/
  generate_perf.cpp, benchmark_np.cpp
tests/
  unit/      csv/, non-preemptive/, preemptive/, leader/
  integration/
  gui/
  data/      input CSVs + ket_qua/ (expected outputs)
  third_party/doctest.h
docs/
  api.md, algorithms.md, conventions.md, plan.md,
  NonPreemptive.md, Preemptive.md, GUI.md, csv-handler.md,
  test-cases/
input/       sample.csv, sample_20.csv (perf_*.csv gitignored)
output/      5 sample result CSVs
report/      non-preemptive-optimization.md, report.md
```

Appendix C — Invariants

The following invariants are enforced by both schedulers and verified by the test suite:

#	Invariant
I1	`ScheduleResult.processes` preserves input order
I2	Gantt covers `[0, max(CT)]` with no gaps and no overlaps
I3	No two adjacent Gantt segments have the same PID
I4	IDLE is represented by `pid = -1`
I5	Schedulers do not fill `average*` fields; `Metrics` does
I6	Empty input → empty `ScheduleResult`, no crash
I7	`WT = TAT − BT`, `TAT = CT − AT`, `RT = first_start − AT`
I8	No context switch overhead, no I/O modeled